International Journal of Information Management Data Insights 2 (2022) 100083

Contents lists available at ScienceDirect

International Journal of Information Management Data Insights

journal homepage: www.elsevier.com/locate/jjimei

Success attributes of business leaders from information technology industry: Evidence from India

Hemant Tiwari, Ramakrishnan Raman*

Symbiosis Institute of Business Management, Pune, India & Symbiosis International (Deemed University), Pune, India

ARTICLE INFO

Keywords:
Leadership
Success attributes
IT industry
Analytical
relational
Personal
Strategic
Functional competencies

ABSTRACT

Leadership has long been associated with the success or failure of an organization. A good leader has certain defined attributes that contribute to the desirable or undesirable outcomes for an organization. Several studies have been conducted that have identified over 100 different attributes associated with effective leadership. Through this paper, the success attributes of business leaders in the IT industry in India are studied. The study made use of qualitative research approach wherein in-depth interviews were conducted with 24 leaders working in different IT organizations in Mumbai. Using thematic analysis, four segments of attributes i.e., analytical, personal, relational and strategic were identified, which contribute to the functional competencies of the leaders. The top four success attributes of Indian business leaders from the IT industry were integrity, identifying the right talent, job knowledge and social and self-awareness, which catalyzed organizational change as the dominant functional competency of the leaders.

1. Introduction

Successful organizations have successful leaders as their backbone, infusing power and intellect within organizational processes. A successful and effective leader exhibits leadership skills by leading the organization, the people and also self (Koohang et al., 2017). Leaders are different from managers, as they do not just manage or direct the team instead, they tend to be the ones that guide teams and help them achieve success. Leaders are therefore essential, as they contribute to organizational performance by promoting knowledge management processes and elevating trust among people (Koohang et al., 2017). The success of the leaders is defined by the skills and knowledge they possess and if they can sharpen the necessary attributes with time and practice (Maedche, 2016).

There have been several studies conducted on leadership and the attributes possessed by successful leaders. These studies have helped in distinguishing the successful from the unsuccessful leaders. Some of the predominant leadership attributes extracted in previous research works include cognitive capacities, social capacities, personality, motives, core beliefs, knowledge and skills. There are over 70 sub-categories for the leadership attributes (Müller & Turner, 2010; Zaccaro, 2007). However, it is crucial to understand that leaders for all industries do not necessarily require the same attributes to be successful. The different industries have different functioning processes, which is why there is a need for leaders to be skilled as per the industry requirements.

IT industry is one of the most thriving industries in today's digital era. The technological advances have created a significant gap in the current capabilities of leaders and the rapidly evolving roles of human resources (Whysall et al., 2019). Leaders are required to possess the ability to accept and adopt change while practicing systems-thinking orientation and guiding the team to achieve organizational objectives. The specific traits of the IT leaders that make them successful may extend beyond the communication skills to being the ability to understand and integrate technology. However, it is to be noted that the ability to succeed even in the IT industry requires a leader to be able to understand the job and be prepared for it, as an organization's success or failure is dependent on the leader's abilities (Al-Ghazali, 2020).

The paper focuses on the IT industry and the attributes required by leaders to be successful in this particular industry. To understand the leader attributes contributing to the success of leaders in the IT industry, the study begins with reviewing the existing literature on the subject. Consequently, the research gaps are identified, and the research question is framed. The methodology used for conducting the study i.e., qualitative analysis is integrated, followed by thematic analysis and identification of the attributes of successful leaders in the Indian IT industry.

The paper is segmented into six sections. The first section introduces the study by providing a brief background on the importance of leadership and different attributes exhibited by the leaders. In the second section, the existing literature on the subject is reviewed leading to the development of research questions based on identified research gap. The research design and interview protocol are explained in the third sec-

* Corresponding author.
E-mail address: director@sibmpune.edu.in (R. Raman).

https://doi.org/10.1016/j.jjimei.2022.100083
Received 24 April 2022; Received in revised form 18 May 2022; Accepted 18 May 2022
2667-0968/© 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

tion, which is followed by data analysis in the fourth segment of the paper. The findings of the study are discussed in the fifth section along with research implications and limitations followed by a concluding paragraph on the key points emerging from the undertaken research,

2. Review of literature

Leadership is defined in the simplest words as the practice or action that involves leading a group of people or a complete organization. Defining leadership considering all its spheres and conceptual aspects is not an easy aspect. However, some of the authors have made an effort to come up with a reasonable definition. For instance, Northouse defined leadership as the process wherein an individual i.e., a leader influences a group of people i.e., followers to perform certain activities to achieve a shared goal (Northouse, 2021). In another study by Bass, leadership is referred to as the method using which the multidisciplinary members of an organization can be brought together to provide quality outcomes through optimum utilization of the available resources (Bass, 2019).

Numerous studies have been focused on the concept of leadership, its importance and the different leadership styles along with their advantages and disadvantages (Al-Ghazali, 2020; Chan & Chan, 2005; Dwiri & Okatan, 2021; Limsila & Ogunlana, 2008; Zaccaro, 2007). The scholars have also studied the differences in the leadership styles adopted by different genders and the subsequent outcomes (Madanchian et al., 2017; Mroz et al., 2018; Wang et al., 2013). The focus of studies on the purpose served by leadership and its impact on organizational performance has gained momentum in the constantly changing global business environment. The constant need to adapt and manage change while re-orienting organizational policies with updated goals generate the need for effective leadership to guide and direct the team towards the attainment of organizational objectives. Effective leadership can also be a significant factor in bringing about a positive change that helps the organizations to improve and at the same time keep innovating in the current business environment (Hao & Yazdanifard, 2015).

Leaders have a certain way of contributing to the organizational goals, managing knowledge transfer within the organization among employees, evaluating how organizational challenges can be overcome and getting the employees to trust the motives of the organization (Koohang et al., 2017). Only when effective leadership skills are exhibited by a leader, the organizational success can be ascertained amid the positive impact of the leader's desirable traits. Madanchain et al. indicated that the effectiveness of a leader is easily measured through the achieved outcomes (Madanchian et al., 2017). Thus, good leadership is identified by the practice wherein a leader can meet the goals set by the organization while keeping the employees onboard and not taking any ethical shortcuts or lowering the expectations from the team (Karp, 2020). Further, Ortenblad et al. have also indicated that good leadership is exhibited when there are no adverse consequences of the decisions made and actions taken (Örtenblad et al., 2016).

Each leader has a specific set of attributes that define the leadership skills and qualities exhibited by them. The recent studies focusing on the attributes of successful leaders led to the development of the categories and sub-categories for the attributes of the leaders, which have been realized as cognitive capacities (intelligence, creative thinking, problem-solving, etc.), social capacities (social Intelligence, self-monitoring, emotional regulation, etc.), personality (big five, optimism, adaptability, charisma etc.), motives (dominance, power, achievement, drive, energy etc.), core beliefs / self-evaluation (locus of control, self-efficacy etc.), knowledge and skills (knowledge of business) and others (honesty/integrity, nurturance, humility, courage etc.) (Al-Ghazali, 2020; Bass, 2019; Maedche, 2016; Müller & Turner, 2010; Zaccaro, 2007).

With the emergence of big data and AI, leaders must be prepared to shoulder the responsibilities in the transforming organizations, proactively adopting the new technology. There have been recent studies focused on the transformative and participative traits of leaders based on their importance in the IT industry and the associated latest technology (Kar et al., 2020; Kumar et al., 2021; Kushwaha et al., 2021; Votto et al., 2021). The studies have concluded that the digital innovations have led to value co-creation and increased the expectations from the leaders to be able to communicate and present the organizational goals and objectives in a manner that coincides with the technological advances (Adikari et al., 2021; Stone & Can, 2021). Moreover, the changing times and business environment, like COVID-19 have introduced changes in the leadership traits expected from the leaders of today's organizations (Koch et al., 2021).

Through the review of the existing literature on the subject, several studies on the leadership attributes of successful leaders have been found. The identified attributes are generalizable and can be associated with leaders from across different industries. However, there are only limited studies that have focused their attention on the attributes of IT industry leaders and negligible that have studied Indian leaders from the IT industry. Additionally, the existing studies do not have segregation of success attributes for leaders holding different positions in the same industry. Hence, a research gap is identified, and the researchers use this gap as an opportunity to study the domain-specific success attributes of IT industry leaders focusing on the Indian business leaders (COOs, CFOs, CMOs and CEOs) from the industry.

Based on the identified research gap, the study will answer the following research question: *What are the key leadership attributes of successful Indian business leaders in the IT industry?*

The underlying research questions that will be answered through the study outcomes are as follows:

- What are the top common general leadership attributes that help leaders achieve success?
- What are the domain-specific leadership success attributes exhibited by COOs, CFOs, CMOs and CEOs in the IT industry?
- Is there any similarity in the leadership success attributes exhibited by COOs, CFOs, CMOs and CEOs in the IT industry?

3. Research methodology

3.1. Design

The design of the undertaken study was influenced by the research objective of identifying the success attributes of IT industry Indian business leaders. To maintain flexibility while exploring attributes of the leaders from the untapped industry, the exploratory research design was integrated (Ohemeng et al., 2018). It is the qualitative nature of research that allows a researcher to gain deeper insights into a topic through the integration of one-on-one dialogues. A semi-structured open-ended trigger questions-based questionnaire was developed to collect data from Indian business leaders over two years through in-depth interviews, in a recorded form, which was then transcribed for further analysis. Although, empirical methods are considered to be the dominant approach of analyzing the data, but the open-ended exploration fetches deep insight into hidden phenomena (Berente et al., 2019).

3.2. Participants

The face-to-face and telephonic in-depth interviews were conducted with 24 executives from various IT organizations in Mumbai extracted from the list of Fortune 500 companies. Fortune India consists of the companies recognized for their top performance based on latest sales and gross revenue (Fortune India: Business News, Strategy, Finance and Corporate Insight 2022). Since the study was focused on identifying the attributes of successful leaders, the top performing organizations were considered ideal for the study. As there were only 24 IT companies in the Fortune 500 list, therefore the data was saturated with 24 leaders when a leader was selected from each company. Data collected through interviews was huge and it was necessary to remove some data that was not meaningful based on the research questions Such data cleaning

approaches can reduce internal and external validity problems and can help in the greater reproducibility of findings (Kar & Dwivedi, 2020).

The sample comprised of five female leaders and 19 male leaders, which can be attributed to the under representation of women at executive levels. The survey conducted by Mercer highlighted that representation of women accounted for only 4-8 percent at executive levels and 12-17 percent at managerial levels (Women under-represented in leadership positions across industries: Survey 2022). From the sample, nine business leaders had profit and loss responsibility of the company i.e., CEOs, seven business leaders handled the administrative responsibilities i.e., COOs, six business leaders were from the finance department i.e., CFOs and two business leaders were from the marketing department overlooking the marketing functions i.e., CMOs. The age of leaders varied between 40 to 60 years old. The participants were selected using the non-probability purposive sampling method, as the participants were being selected for a purpose (Etikan, 2016).

3.3. Interview protocol

Data were collected over two years. This was owing to the scheduling challenges for the time of the senior industry leaders. This study was conducted using a semi-structured interview protocol. 60% of the interviews were conducted face-to-face and audio recorded with permission. 40% of the interviews were conducted over the telephonic discussion and the call/interview was recorded with permission.

The set of semi-structured questions was shared with participants in advance for their reference. This was to enable them to structure their thoughts and recollect insights from their vast experience. The interviews were then conducted by introduction, stating the purpose and topic of discussion. At the same time, permission to record the conversation was also taken from each participant. Each interview lasted anywhere between 60 to 90 min. In three instances, it was not possible to record the interview and the hand-written notes were taken for data collection. The selection of the questions was done to generate insights on key dimensions of the business leaders for the organizations. The semi-structured interview reference reckoner used by the interviewee had 22 questions but not necessarily all questions were asked to all the interviewees. The set of questions asked during the interview was decided on the spot during the interview depending on the responses from the participants, already generated insights, time availability and flow of the interview.

Overall, the importance of conducting interviews with the business leaders in the targeted industry lies in their unique experiences and insights about what makes them successful and then drawing commonality from their experiences (Gritzo et al., 2017). The questions in the interview began with the general views of participating leaders about leadership. Some of the questions that they were asked were:

- What in your view is leadership?
- What are the top 3 attributes of a leader who had a significant influence on you?
- What are some qualities you correlate with leaders who demonstrate success?
- Whom do you call successful leaders?

The questions also ranged from managing culture to managing change and the attributes needed in such situations. The leaders were asked about their ability to manage work-life balance, attributes needed to work successfully within constraints, the importance of coaching and grooming other leaders, attributes needed to overcome failures and obstacles and the attributes needed to manage difficult stakeholders and communication.

Overall, the aim was to generate insights into the attributes each leader would have. Consequently, interesting and insightful anecdotes were obtained during the interview process.

Fig. 1. Word cloud of the most frequently used words by the participants.

4. Data analysis

After the completion of the data collection process and attainment of transcription of interviews, the data was transformed and modelled into useful information to extract conclusions. The data analysis proceeded with word cloud development and grounded theory application for the undertaken study.

4.1. Word cloud

The integration of the data visualization process became the first step in analyzing the data about the success attributes of Indian business leaders in the IT industry. The application of the visualization technique in the evaluation of qualitative data is helpful for the researcher, as it makes the presentation of most integral elements to be considered, simpler and clear (Henderson & Segal, 2013). Through the word cloud method, the lengthy text-based data was evaluated, and insights were gained by comprehending the data. It can be observed in Fig. 1 that several words are depicted in different sizes along with the differences in some words being bolder than others. The difference in the size and intensity of the appearance of words is based on their repeated occurrence in the transcribed interviews that were analyzed.

From the figure, it can be interpreted that people are the most frequently used word followed by leader, organization, communication, change and team experience. It leads to the development of an understanding that the transcripts mentioned people numerous times and the participants talked about the people in the organization and how the work is done through communication. The focus on change brings forth the understanding that the leadership skills executed by the participants revolved around the adjustment to the changes that an organization goes through. Some of the other words that have been highlighted through the word cloud include management, influence, challenge, vision customer, expectation and culture.

It further generates the understanding that the participants were aware of the impact that they have on the organizational culture and the way their vision takes the organization forward. The ability of the organization to successfully manage its customers is also brought forth in the discussion and interview collected from the participants. The investigation of these patterns that emerge from the words of phrases is generally helpful for the researcher to identify common themes in the feedback given by the participants and resultantly understand the concepts hidden within (DePaolo & Wilkinson, 2014). As word cloud helped in identifying the most frequently used words in the transcripts, the technique can be identified as the basis of the coding process.

4.2. Thematic analysis

Once the most frequently appearing words were identified using word cloud, the data was further analyzed using grounded theory. Grounded theory allows the researcher to understand those phenomena that cannot be explained using existing theory (DePaolo & Wilkin-

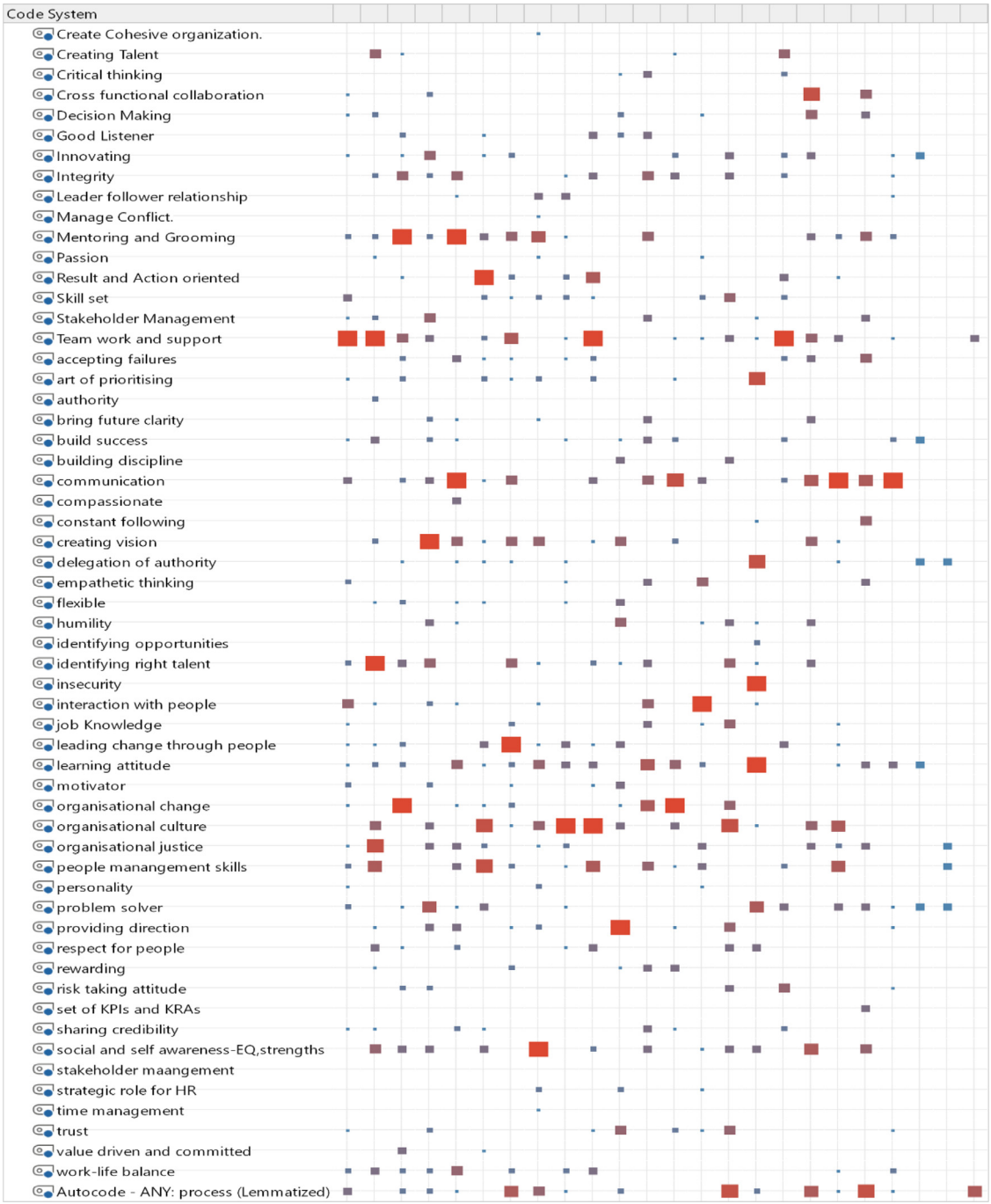

Fig. 2. MAXQDA matrix.

son, 2014). The application of grounded theory will allow the researcher to critically review participant response and possibly develop a theory. Thematic analysis is the method that uses systematic identification and organization of the data set leading to insights into the patterns that emerge from the collected data. Thematic analysis is advantageous in analyzing the large data set, as it makes sense of the overall data that has been collected from the respondents and does not draw inferences from standalone data. In the process of conducting thematic analysis, codes are generated based on the common phrases and words which can be associated with a defined idea. To be able to generate these codes and categorize the transcript data MAXQDA software was used. The results obtained from the application of the software can be seen in Fig. 2.

The application of the MAXQDA software is most feasible in cases when there is a large dataset like interview transcripts. The matrix obtained using the software shows that several phrases appeared repeatedly in the analyzed transcripts. The codes that show large red boxes indicate the frequency of the repetition in the transcript of the interview from one participant.

Organizational change and organizational culture were identified as the phrases repeated most frequently in the analyzed interview transcripts. The other phrases that were found to be repeated frequently included mentoring and grooming, result and action-oriented, teamwork and support, communication, creating a vision, identifying the right talent, learning attitude, people management skills and social and self-awareness.

Table 1
Frequency of commonly occurring words in analyzed documents.

	Frequency	Percentage	Percentage (valid)
Communication	15	62.50	68.18
Mentoring and Grooming	14	58.33	63.64
Social and Self-Awareness-EQ Strengths	12	50.00	54.55
Build Success	11	45.83	50.00
Innovating	11	45.83	50.00
Integrity	11	45.83	50.00
Work-Life Balance	10	41.67	45.45
Organizational Change	9	37.50	40.91
Result and Action-oriented	7	29.17	31.82
Job Knowledge	6	25.00	27.27
Decision Making	6	25.00	27.27
Good Listener	5	20.83	22.73
Creating Talent	4	16.67	18.18
Critical thinking	3	12.50	13.64
identifying opportunities	1	4.17	4.55
Documents with code(s)	22	91.67	100.00
Documents without code(s)	2	8.33	-
Analyzed documents	24	100.00	-

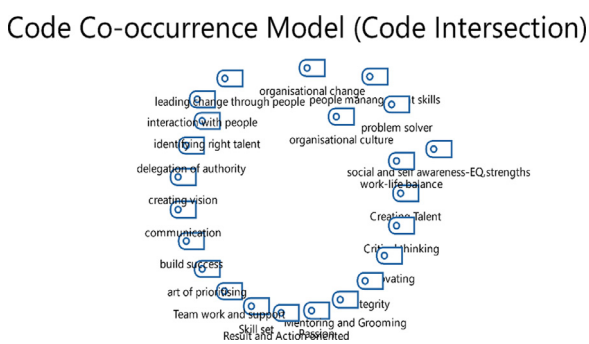

Fig. 3. Prominent codes obtained through document analysis.

Thus, the matrix suggests that while some attributes are more prominent than others, there are several attributes, which can be identified as the factors of success for leaders in the IT industry.

Further analysis of the data was conducted to understand how some attributes occur automatically when a certain defined attribute is found in the leader. Fig. 3 presents the code co-occurrence model which throws light on the prominent codes that were obtained to document analysis of the coexistence of two different attributes at the same time.

The application of the code cooccurrence model or code in the intersection model helped in the identification of prominent codes that were overlapping in the coded segments. It presents the frequency of the occurrence of certain codes in presence of the other identified and defined codes.

The frequency of commonly occurring words in the analyzed transcripts was refined further. Table 1 presents that there were in all 15 phrases that occurred most frequently among all the other identified and prominent codes.

The table highlights that communication was the most frequently repeated word and identifying opportunities was the word used only once. It is also seen that 22 documents had codes from the 24 analyzed documents. Two of the documents that were analyzed did not contain any code.

The generation of the initial codes helps in the identification and labelling of the part of data that is relevant to the question. These codes then help the researcher to capture something about the data which can help answer the research question and therefore a shift from courts to themes is observed (Castleberry & Nolen, 2018). Once the categories of attributes were obtained, they were further coded based on their relativity. In the undertaken research study, the focus is on the success attributes, therefore, attributes related to themes were generated. Table 2 presents the way the categories or the attributes were segmented, and the relevant themes were generated.

Through this segmentation of the attributes using thematic analysis the generated themes were recognized as analytical attributes, personal attributes, relational attributes, strategic attributes and deliverables. While critical thinking and decision-making formed the part of analytical attributes; it was social and self-awareness, integrity and a good listener that was added to the personal attributes category. Communication and job knowledge was categorized as relational attributes and strategic attributes consisted of mentoring and grooming, innovating, identifying opportunities and creating talent. The fifth theme i.e., deliverables were understood to be the outcomes that were achievable by the leaders who possessed the combination of two or more attributes from different segments. These deliverables can also be referred to as the functional competencies of the leaders possessing certain defined attributes. The functional competencies included work-life balance, organizational change, result and action-oriented and building success.

To further analyze the similarities and differences in the attributes of different business leaders in the IT industry two-case model was developed. Fig. 4 presents the dominant attributes that were found to be common for both CEOs and COOs of the IT industry. The figure shows different attributes that were essential for CEOs and COOs as well as the attributes that were common for both.

It can be interpreted from the figure that CEOs and COOs of the industry exhibited integrity and social and self-awareness as the personal attributes, identifying and creating the right talent as the strategic attributes and job knowledge as the relational attribute. These attributes form the three dominant themes extracted using thematical analysis formed the basis of the IT industry's COOs' and CEOs' functional competency i.e., organizational change.

Conclusively, through the thematic analysis of the transcripts, it was found that there are certain defined attributes of successful leaders in the IT industry. However, it was also observed that different business leaders handling different departments in the same industry may or may not have similar attributes contributing to their success.

5. Discussions

The analysis of the transcripts generated from the recorded in-depth interviews with 24 leaders led to the identification of 15 attributes that were repeated constantly by all the participants. A discussion on these 15 attributes that were segmented into five different themes is presented in the following sections.

Table 2
Themes are generated based on segmented categories (attributes).

Analytical Attributes	Personal Attributes	Relational Attributes	Strategic Attributes	Deliverables
Critical Thinking Decision Making	Social and Self-Awareness – EQ Strengths Integrity Good Listener	Communication Job Knowledge	Mentoring and Grooming Innovating Identifying Opportunities Creating talent	Work-Life Balance Organizational Change Result and Action Oriented Build Success

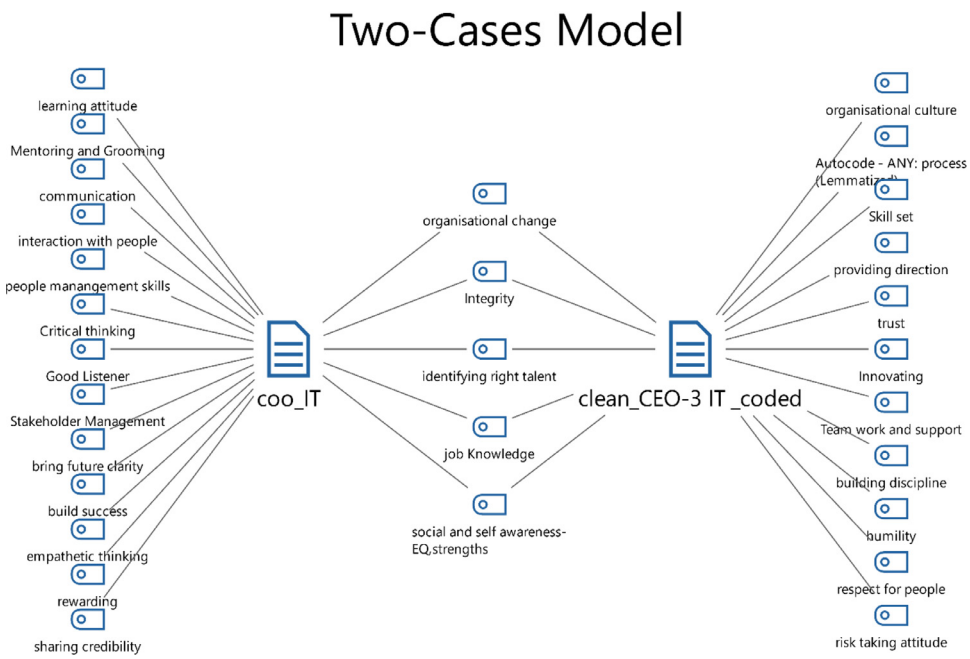

Fig. 4. Dominant attributes common to COO and CEO.

5.1. Analytical attributes

Critical thinking: 12% of the participating leaders highlighted that critical thinking plays a crucial role in the effective management of the organizational process. Zivkovic mentions that critical thinking i.e., the ability to analyze the available facts and base the decision on them, is a necessary skill for leaders (ŽivkoviL, 2016). The leaders ought to observe their surroundings and scrutinize the arguments presented to make a judgment. In the organizational processes, these judgments can have an impact on the overall organizational growth and may lead to the success or failure of the implemented strategy (Chikeleze et al., 2018). Critical thinking further gains importance in the 21st century amid a constantly changing business environment, therefore, making it one of the most important success attributes.

Decision making: The process of decision-making is omnipresent within the organization and 25% of the participants felt that decision-making is one of the most important attributes that can play an influential role in their success. The leaders are on the constant trail of making decisions for the organization based on the organizational goals, resources and capabilities. Strong decision-making ability is also essential to ensure that everything is planned and executed within the desired timeline (Aij & Teunissen, 2017). Leaders are responsible for the decisions they make, as their decisions can influence the success of the strategic implementations or can lead the organization to adverse outcomes. Therefore, leaders need to use their critical thinking skills to analyze the possible consequences of their decisions while evaluating the options and navigating risks (Ejimabo, 2015).

5.2. Personal attributes

Social and self-awareness - EQ strengths: 50% of the participants talked about the importance of emotional intelligence in their routine activities and functioning as a leader. Gomez-Leal et al. concluded in their work that the emotional intelligence of individuals is linked with the combination of their cognitive skills with their emotions facilitating the capacity of pursuing empathetic communication and likes (Gómez-Leal et al., 2021; Kim & Kim, 2017). The success attributes of business leaders in the IT industry of India were identified as social awareness and self-awareness, which indicated that the ability to understand social situations and conflicts as well as the power to look within and overcome the limitations contribute to the progress made by the leaders. It is a success attribute shared by the Indian IT business leaders serving as COOs as well as those as CEOs.

Integrity: Integrity, as quoted by one of the leaders, is the sum of being honest, trustworthy and reliable. In other leaders' words, it is all about practicing what one preaches and owning up to the mistakes. Integrity adds to the moral values executed by the leaders, which affects the way their team looks up to them (Krylova et al., 2017). It has close relation with the attribute "trust", which is important to influence the team toward organizational vision and short-term tactical goals. When leaders act with integrity, they win the trust of their people. Integrity and trust form the solid building blocks for a leader-follower relationship. The importance of integrity is further understood through the fact that both COOs and CEOs of the IT industry, among the participants, identified it as a success attribute.

Good listener: To be a good leader, being a good listener is one of the key personal attributes. 20% of leaders agreed to the fact that listening skills contribute to the overall success they achieve in their position. The leaders are expected to ask a lot of questions to their team members, customers and also the top management. However, it is not just the process of questioning that is crucial and instead, listening is the key aspect of interaction (Van Quaquebeke & Felps, 2018). Additionally, when the employees feel that their leader is listening to them, there is a significant reduction in emotional exhaustion and turnover intention (Lloyd et al.,

2014). Thus, a good listener among leaders is the one who encourages positive organizational citizenship behavior among employees.

5.3. Relational attributes

Communication: Communication was rated as the top success attributes by leaders. Almost 68% of the leaders highlighted it as the key attribute for success. From managing critical and difficult communication to managing key stakeholder communication; from communicating vision clearly to communicating to build and drive culture, leaders succeed when they are able to communicate clearly and effectively (Ruben & Gigliotti, 2016). It also came out as an attribute which cuts across in most of the situations and has a maximum impact on the success of leaders. Communication is therefore the key success attribute, which influences the overall organizational progress and performance through transparency among different levels of management (Luu, 2021).

Job knowledge: Leaders are the individuals that are always looked upon by the team members. 25% of the participants agreed that job knowledge contributed to their success as a leader. The attribute of job knowledge indicates that the leaders are expected to know what they are doing and should have adequate knowledge of the tasks they are to perform. Additionally, Taylor et al. have indicated that the leaders are responsible for knowledge sharing to help the team pursue shared organizational goals. The ability of the leaders to transfer adequate knowledge as per the requirement of the tasks and the need of the employees helps in development of positive relationships in the organizations (Arora & Rangnekar, 2015).

5.4. Strategic attributes

Mentoring and grooming: Arora and Rangnekar concluded in their studies that the leaders help in the improvement of organizational outcome through constant mentoring and grooming of the employees (Holmes & Carr, 2017). When leaders act as mentors, they upskill the employees (mentees) and nurture them by encouraging them to learn and grow as per the demands of the business environment. In turn, the organization is benefitted as the employees feel motivated and valued as well as confident to deliver their best in the job responsibilities. The progress made by the employees in their career helps the leaders under the impact of reverse mentoring. The motivating language used by leaders and encouraging engagement in the routine activities make employees have regard towards them, adding to their success (Perri et al., 2019).

Innovating: Almost 46% participants agreed that innovation is the need of the hour and their ability to innovate makes them successful in their respective positions. The constantly changing business environment, increasing demands of the consumers and prevailing competition in the industry can also be managed through constant innovations. The stakeholders also push the leaders to innovate so that a competitive edge can be maintained within the industry (McElheran, 2015). Therefore, it can be understood that leaders who have innovative capabilities are more successful in contributing to the progress of the organization. McElheran has indicated that the leaders with innovative capabilities have the ability to adopt the required changes in the business processes and keep the organizational goals aligned with changing market needs (Qu et al., 2015).

Identifying opportunities: Identifying opportunities is a key leadership skill but in the interviews only one participant highlighted it as an attribute contributing to success. There are endless opportunities surrounding today's businesses and a good leader should be able to identify those and use those favorable circumstances to strengthen the organization's position in the market. Furthermore, being able to identify opportunities gives the leaders an added advantage when it comes to getting the best from the employees (Singh, 2015). Being able to understand and recognize opportunity in the suggestions made by team members further accelerate the growth of business and make leaders successful. Even identifying the opportunity to upskill and reskill helps leaders stay ahead in the competition.

Creating talent: A leader is responsible for understanding the objectives of the organization and evaluate the changing requirements as per the business needs. The leader shoulders the responsibility to creating a team that is competent and capable of taking on the new challenges. In doing so, a leader can provide training and upskilling opportunities, which boost the talent of the individuals (Masenya, 2022). The ability to cultivate talent in the employees makes the process of adopting changes simpler and more convenient. Moreover, in the rapidly changing environment, it becomes crucial for the leaders to ensure that the employees are able to adopt to the changes, which is possible through the creation of talent (Shet et al., 2017). The COOs and CEOs of Indian IT industry confirmed that creating talent is one success attribute common to both types of business leaders.

5.5. Deliverables

The fifth theme that was identified through the segmentation of categories of attributes did not merely define the traits that leaders have. Instead, it led to the identification of the outcome of the attributes that were identified and extracted from the interview transcripts. They are categorized as deliverables, as the success attributes of the leaders enable them to deliver certain outcomes. Shet et al. have indicated in their study that the ability of the professionals to deliver something forms the capability of functional attributes that they possess and catalyzes the process of their recruitment (Mattock, 2015). Therefore, the fifth themes can also be referred to as functional attributes.

Work-life balance: Work-life balance is one of the most important phenomena observed in today's era. Almost 42% of the participants focused on work-life balance confirming its importance. Most of the employees feel that they have a significant disturbance in their work-life balance and often it is the lack of effective leadership, which disturbs this balance. When a leader is able to manage organizational processes and define working conditions in such a manner that a balance is maintained, employees feel content (Øygarden et al., 2020). Therefore, leaders that are able to deliver work-life balance to self as well as the team are preferred and given high regard.

Organizational change: The only deliverable or functional attribute that was found to be common for both COOs and CEOs among study participants was organizational change. Holten and Brenner concluded in their work that the constantly changing business environment requires the organizations to undergo rapid changes. However, it becomes difficult to manage the organization's environment and the employees tend to resist the change. Only in the presence of a leader who is effective and efficient can the organizational change proceed smoothly (Olson & Simerson, 2015). Therefore, a leader who has the capability of communicating with the team and while mentoring them to be change ready, can successfully deliver organizational change. It proves that success attributes of a leader catalyze certain outcome, which can be segmented as deliverables or functional attributes.

Result and action oriented: The leaders possessing job knowledge and having an eye for talent can deliver strategies that are result oriented. Moreover, the capacity of identifying opportunities lead to actions that are favorable for the organization. Being focused on results make the leaders able to drive change (Heskett et al., 2015) and their actions catalyze breakthroughs that empower the employees as well as the organization (Holten & Brenner, 2015).

Build success: 46% participants identified that their attributes help them in leading the organization towards success. The ability of the leaders to identify opportunities, maintain social and self-awareness and also create talent contributes to the potential of the organization and delivers successful outcomes to organizational objectives. Success is associated not just with that of the organization and extends beyond to the career success of the employees as well as leaders (Ruben & Gigliotti, 2016).

Fig. 5. Leadership attribute framework for IT industry.

Through the findings and discussions of the research study a leadership attribute framework can be proposed which brings out the interrelations ship between the attributes and functional competencies of a leader in the Indian IT industry. Fig. 5 presents the proposed framework with identified attributes and their ability to make the leaders deliver defined objectives.

The leadership attribute framework proposed indicates that the attributes are interrelated, and the combination of these attributes helps the business leaders in the Indian IT industry to deliver what is expected of them.

5.6. Implications

The results obtained from the study have theoretical as well as managerial implications. On the basis of the defined success attributes of business leaders in the IT industry, it can be theorized that each industry possibly has specific requirements from its leaders. Therefore, a generalized understanding of the success attributes of leaders cannot help in the development of an understanding of successful leaders and their attributes from a defined or domain-specific perspective. The theoretical understanding that certain attributes of leaders catalyze their success thus needs to be refined and domain-specific studies are required to help gain better insights on the subject.

Secondly, when it comes to managerial implications, the identification of the required attributes in successful leaders of specific industry like IT industry in India extracted in this paper can help in the recruitment of suitable candidates. The top management can assess the candidates for the domain specific attributes and select the one who emerges as the right fit for the position. It would further be useful for the business leaders to understand these specific traits that they need to either possess or develop so as to become successful in their position and contribute to organizational change.

The research implications are also associated with the fact that the traits or attributes of leaders can be categorized into four different segments i.e., analytical, personal, relational and strategic based on their impact on leadership processes. Finally, the results obtained from the research imply that the attributes of successful leaders in the industry lead to certain defined functional competencies of leaders which are associated with their effectiveness and positive contribution towards the organizational growth and development.

5.7. Limitations and future research directions

The research outcomes help in the development of an understanding that these success attributes of leaders make them functionally competent. It is identified and concluded that all the attributes of successful leaders can be segmented into four themes i.e., analytical, personal, relational and strategic aspects. These attributes either individually or combined can help leaders deliver certain specific outcomes. However, the undertaken study is limited to the Indian IT industry and results obtained are domain specific. Therefore, future studies can focus on developing a comparative analysis of attributes specific to different domains or industries within India. For example, a study can be undertaken to understand differences and similarities in the success attributes of business leaders hailing from a product industry (IT) and that from the service industry (BFSI). Additionally, future research can be conducted to bring out the similarities and differences in specific attributes of leaders serving in different departments within an organization. For instance, the specific success attributes of CEOs maybe different from those of CMOs, CFOs, CHROs and other similar top business leaders.

Moreover, in the literature review undertaken for the current study a slight indication of differences in leadership attributes based on the gender of leaders was identified. Subsequently, it can become the basis of future research direction. The study can expand on why male leaders are preferred over female leaders and if there is a relationship between the attributes of leaders leading to their success based on their genders (Taylor et al., 2015). Through the future research focus on the gender-based identification of success attributes, the leadership attribute frame-

work can further be evolved and upgraded to show how the attributes and functional competencies can be determined based on gender differences. A possible focus on the differences in success attributes based on their leadership styles can also be studied to evaluate the relationship between leadership style and leader attributes (Abbas & Ali, 2021; Taylor et al., 2015).

6. Conclusion

To conclude, the study focused on the analysis of the attributes of successful leaders in the Indian IT industry brought forth the four main categories of leader characteristics. The study made use of qualitative approach and analyzed the interview transcripts collected from 24 leaders from 24 IT industries listed under Fortune 500 companies. The grounded theory approach led to the generation of five themes of which four themes defined analytical, personal, relational and strategic attributes and the fifth theme explained the functional competence of leaders emerging from the 15 coded attributes. The study contributes to the understanding about industrial requirements from leaders and the possible application of attribute-specific recruitment direction. It also identifies an opportunity for future researchers to compare the industry specific attributes and identify gender-based differences in attributes of successful leaders.

Disclodure statement

There is no conflict of interest.

CRediT authorship contribution statement

Hemant Tiwari: Conceptualization, Writing – original draft. **Ramakrishnan Raman:** Conceptualization, Writing – review & editing.

References

Koohang, A., Paliszkiewicz, J., & Goluchowski, J. (2017). The impact of leadership on trust, knowledge management, and organizational performance. *Industrial Management & Data Systems, 117*(3), 521–537. 10.1108/imds-02-2016-0072.

Maedche, A. (2016). Interview with Michael Nilles on "what makes leaders successful in the age of the digital transformation? *Business & Information Systems Engineering, 58*(4), 287–289. 10.1007/s12599-016-0437-1.

Müller, R., & Turner, J. (2010). Attitudes and leadership competences for project success. *Baltic Journal Of Management, 5*(3), 307–329. 10.1108/17465261011079730.

Zaccaro, S. (2007). Trait-based perspectives of leadership. *American Psychologist, 62*(1), 6–16. 10.1037/0003-066x.62.1.6.

Whysall, Z., Owtram, M., & Brittain, S. (2019). The new talent management challenges of Industry 4.0. *Journal Of Management Development, 38*(2), 118–129. 10.1108/jmd-06-2018-0181.

Al-Ghazali, B. (2020). Transformational leadership, career adaptability, job embeddedness and perceived career success–A serial mediation model. *Leadership & Organization Development Journal, 41*(8), 993–1013. 10.1108/lodj-10-2019-0455.

Northouse, P. (2021). *Leadership* (9th ed.). SAGE Publications.

Bass, B. (2019). What is leadership? In *Leadership in surgery* (pp. 1–10). 10.1007/978-3-030-19854-1_1.

Chan, A., & Chan, E. (2005). Impact of perceived leadership styles on work outcomes–Case of building professionals. *Journal Of Construction Engineering And Management, 131*(4), 413–422. 10.1061/(asce)0733-9364(2005)131:4(413.

Dwiri, B., & Okatan, K. (2021). The impact of gender on leadership styles and leadership effectiveness. *International Journal Of Science And Research (IJSR), 10*(1), 1419–1434. 10.21275/sr21126183926.

Limsila, K., & Ogunlana, S. (2008). Performance and leadership outcome correlates of leadership styles and subordinate commitment. *Engineering, Construction And Architectural Management, 15*(2), 164–184. 10.1108/09699980810852682.

Madanchian, M., Hussein, N., Noordin, F., & Taherdoost, H. (2017). Leadership effectiveness measurement and its effect on organization outcomes. *Procedia Engineering, 181*, 1043–1048. 10.1016/j.proeng.2017.02.505.

Mroz, J., Yoerger, M., & Allen, J. (2018). Leadership in workplace meetings–The intersection of leadership styles and follower gender. *Journal Of Leadership & Organizational Studies, 25*(3), 309–322. 10.1177/1548051817750542.

Wang, A., Chiang, J., Tsai, C., Lin, T., & Cheng, B. (2013). Gender makes the difference–The moderating role of leader gender on the relationship between leadership styles and subordinate performance. *Organizational Behavior And Human Decision Processes, 122*(2), 101–113. 10.1016/j.obhdp.2013.06.001.

Hao, M., & Yazdanifard, R. (2015). How effective leadership can facilitate change in organizations through improvement and innovation. *Global Journal Of Management And Business Research, 15*(9), 1–5.

Karp, T. (2020). What do we really mean by good leadership? *Journal Of Values-Based Leadership*. 10.22543/0733.131.1300.

Örtenblad, A., Hong, J., & Snell, R. (2016). Good leadership–A mirage in the desert? *Human Resource Development International, 19*(5), 349–357. 10.1080/13678868.2016.1216736.

Kushwaha, A., Kar, A., & Dwivedi, Y. (2021). Applications of big data in emerging management disciplines–A literature review using text mining. *International Journal Of Information Management Data Insights, 1*(2), Article 100017. 10.1016/j.jjimei.2021.100017.

Votto, A., Valecha, R., Najafirad, P., & Rao, H. (2021). Artificial intelligence in tactical human resource management–A systematic literature review. *International Journal Of Information Management Data Insights, 1*(2), Article 100047. 10.1016/j.jjimei.2021.100047.

Kar, S., Kar, A., & Gupta, M. (2020). Talent scarcity, skill distance and reskilling resistance in emerging digital technologies - Understanding employee behaviour. *International conference in information systems* Retrieved from https://aisel.aisnet.org/icis2020/practice_is_research/practice_is_research/3/ .

Kumar, S., Kar, A., & Ilavarasan, P. (2021). Applications of text mining in services management–A systematic literature review. *International Journal Of Information Management Data Insights, 1*(1), Article 100008. 10.1016/j.jjimei.2021.100008.

Adikari, A., Burnett, D., Sedera, D., de Silva, D., & Alahakoon, D. (2021). Value co-creation for open innovation–An evidence-based study of the data driven paradigm of social media using machine learning. *International Journal Of Information Management Data Insights, 1*(2), Article 100022. 10.1016/j.jjimei.2021.100022.

Stone, J., & Can, S. (2021). Gendered language differences in public communication? The case of municipal tweets. *International Journal Of Information Management Data Insights, 1*(2), Article 100034. 10.1016/j.jjimei.2021.100034.

Koch, J., Plattfaut, R., & Kregel, I. (2021). Looking for talent in times of crisis – The impact of the COVID-19 pandemic on public sector job openings. *International Journal Of Information Management Data Insights, 1*(2), Article 100014. 10.1016/j.jjimei.2021.100014.

Ohemeng, F., Amoako-Asiedu, E., & Obuobisa Darko, T. (2018). The relationship between leadership style and employee performance. *International Journal Of Public Leadership, 14*(4), 274–296. 10.1108/ijpl-06-2017-0025.

Berente, N., Seidel, S., & Safadi, H. (2019). Research commentary–Data-driven computationally intensive theory development. *Information Systems Research, 30*(1), 50–64. 10.1287/isre.2018.0774.

Fortune India: Business News, Strategy, Finance and Corporate Insight. (2022). Retrieved 2 May 2022, from https://www.fortuneindia.com/next-500/company/

Kar, A., & Dwivedi, Y. (2020). Theory building with big data-driven research – Moving away from the "What" towards the "Why. *International Journal Of Information Management, 54*, Article 102205. 10.1016/j.ijinfomgt.2020.102205.

Women under-represented in leadership positions across industries: Survey. (2022). Retrieved 2 May 2022, from https://www.tribuneindia.com/news/jobs-careers/women-under-represented-in-leadership-positions-across-industries-survey-375860

Etikan, I. (2016). Comparison of convenience sampling and purposive sampling. *American Journal Of Theoretical And Applied Statistics, 5*(1), 1. 10.11648/j.ajtas.20160501.11.

Gritzo, L., Fusfeld, A., & Carpenter, D. (2017). Success factors in R&D leadership. *Research-Technology Management, 60*(4), 43–52. 10.1080/08956308.2017.1325683.

Henderson, S., & Segal, E. (2013). Visualizing qualitative data in evaluation research. *New Directions For Evaluation, 2013*(139), 53–71. 10.1002/ev.20067.

DePaolo, C., & Wilkinson, K. (2014). Get your head into the clouds–Using word clouds for analyzing qualitative assessment data. *Techtrends, 58*(3), 38–44. 10.1007/s11528-014-0750-9.

Castleberry, A., & Nolen, A. (2018). Thematic analysis of qualitative research data–Is it as easy as it sounds? *Currents In Pharmacy Teaching And Learning, 10*(6), 807–815. 10.1016/j.cptl.2018.03.019.

ŽivkoviL, S. (2016). A model of critical thinking as an important attribute for success in the 21st century. *Procedia - Social And Behavioral Sciences, 232*, 102–108. 10.1016/j.sbspro.2016.10.034.

Chikeleze, M., Johnson, I., & Gibson, T. (2018). Let's argue–Using debate to teach critical thinking and communication skills to future leaders. *Journal Of Leadership Education, 17*(2), 123–137. 10.12806/v17/i2/a4.

Aij, K., & Teunissen, M. (2017). Lean leadership attributes–A systematic review of the literature. *Journal Of Health Organization And Management, 31*(7/8), 713–729. 10.1108/jhom-12-2016-0245.

Ejimabo, N. (2015). The influence of decision making in organizational leadership and management activities. *Journal Of Entrepreneurship & Organization Management, 04*(02). 10.4172/2169-026x.1000138.

Gómez-Leal, R., Holzer, A., Bradley, C., Fernández-Berrocal, P., & Patti, J. (2021). The relationship between emotional intelligence and leadership in school leaders–A systematic review. *Cambridge Journal Of Education, 52*(1), 1–21. 10.1080/0305764x.2021.1927987.

Kim, H., & Kim, T. (2017). Emotional intelligence and transformational leadership–A review of empirical studies. *Human Resource Development Review, 16*(4), 377–393. 10.1177/1534484317729262.

Krylova, K., Jolly, P., & Phillips, J. (2017). Followers' moral judgments and leaders' integrity-based transgressions–A synthesis of literatures. *The Leadership Quarterly, 28*(1), 195–209. 10.1016/j.leaqua.2016.10.002.

Van Quaquebeke, N., & Felps, W. (2018). Respectful inquiry–A motivational account of leading through asking questions and listening. *Academy Of Management Review, 43*(1), 5–27. 10.5465/amr.2014.0537.

Lloyd, K., Boer, D., Keller, J., & Voelpel, S. (2014). Is my boss really listening to me? The impact of perceived supervisor listening on emotional exhaustion, turnover intention, and organizational citizenship behavior. *Journal Of Business Ethics, 130*(3), 509–524. 10.1007/s10551-014-2242-4.

Ruben, B., & Gigliotti, R. (2016). Communication. *International Journal Of Business Communication, 54*(1), 12–30. 10.1177/2329488416675447.

Luu, T. (2021). Knowledge sharing in the hospitality context–The roles of leader humility, job crafting, and promotion focus. *International Journal Of Hospitality Management, 94*, Article 102848. 10.1016/j.ijhm.2020.102848.

Arora, R., & Rangnekar, S. (2015). The joint effects of personality and supervisory career mentoring in predicting occupational commitment. *Career Development International, 20*(1), 63–80. 10.1108/cdi-12-2014-0156.

Holmes, W., & Carr, M. (2017). Motivating language and self-mentoring–A training program supporting the development of leaders in organizations. *Development And Learning In Organizations: An International Journal, 31*(5), 4–8. 10.1108/dlo-02-2017-0010.

Perri, S., Farrington, T., Johnson, S., & O'Connor, G. (2019). Today's innovation leaders. *Research-Technology Management, 62*(1), 20–29. 10.1080/08956308.2019.1541726.

McElheran, K. (2015). Do market leaders lead in business process innovation? The case(s) of E-business adoption. *Management Science, 61*(6), 1197–1216. 10.1287/mnsc.2014.2020.

Qu, R., Janssen, O., & Shi, K. (2015). Transformational leadership and follower creativity–The mediating role of follower relational identification and the moderating role of leader creativity expectations. *The Leadership Quarterly, 26*(2), 286–299. 10.1016/j.leaqua.2014.12.004.

Singh, M. (2015). Talent management by capacity development of the employees. *SSRN Electronic Journal*. 10.2139/ssrn.2581401.

Masenya, T. (2022). Integrating talent and knowledge management practices in the new normal business environment. *Advances In Human Resources Management And Organizational Development*, 113–144. 10.4018/978-1-7998-8451-4.ch006.

Shet, S., Patil, S., & Chandawarkar, M. (2017). Framework for methodical review of literature on leadership competencies. *Cogent Business & Management, 4*(1), Article 1309123. 10.1080/23311975.2017.1309123.

Mattock, S. (2015). Leadership and work–life balance. *Journal Of Trauma Nursing, 22*(6), 306–307. 10.1097/jtn.0000000000000163.

Øygarden, O., Olsen, E., & Mikkelsen, A. (2020). Changing to improve? Organizational change and change-oriented leadership in hospitals. *Journal Of Health Organization And Management, 34*(6), 687–706. 10.1108/jhom-09-2019-0280.

Olson, A., & Simerson, B. (2015). *Leading with strategic thinking–Four ways effective leaders gain insight*. John Wiley & Sons.

Heskett, J., Sasser, W., & Schlesinger, L. (2015). *What great service leaders know and do* (1st ed.). Oakland: Berrett-Koehler Publishers.

Holten, A., & Brenner, S. (2015). Leadership style and the process of organizational change. *Leadership & Organization Development Journal, 36*(1), 2–16. 10.1108/lodj-11-2012-0155.

Ruben, B., & Gigliotti, R. (2016). Leadership as social influence. *Journal Of Leadership & Organizational Studies, 23*(4), 467–479. 10.1177/1548051816641876.

Taylor, T., Psotka, J., & Legree, P. (2015). Relationships among applications of tacit knowledge and transformational/transactional leader styles. *Leadership & Organization Development Journal, 36*(2), 120–136. 10.1108/lodj-01-2013-0008.

Abbas, M., & Ali, R. (2021). Transformational versus transactional leadership styles and project success–A meta-analytic review. *European Management Journal*. 10.1016/j.emj.2021.10.011.